#9 From a national bioinformatics society to the development of the national bioinformatics ecosystem

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ELIXIR Observer: ROMANIA

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Romanian Society for Bioinformatics

Development of a national bioinformatics society - RSBI

Founding and Evolution

Initial focus
- Training and Networking Events

Curent focus
- Training and Networking Events
- Public Engagement
- Advocacy for Infrastructure and Policy Development

Key initiatives

RoBioinfo Seminars (since 2018)
- Over 30 events across Romania

Beer and Science
- Informal meet-up for scientist interactions

Policy Reform Contributions
- Genomics and Bioinformatics alignment with research realities
- Romanian Government signed the '1+ Million Genomes' declaration (2023)
- Joined Genomic Data Infrastructure and ELIXIR

Data Federation Initiatives

Challenges:
- Need for robust organizational structure

Motivating factors:
- Effective large scale training;
- Federate data, a problem evident during COVID-19;
- Grid of computational resources.

2017
Perspectives of bioinformatics in Romania - national conference, Cluj-Napoca

2018
Start of RoBioinfo Seminars - bioinformatics training

Start of Beer & Science as RoBioinfo satellite public engagement meetings

2019
RSBI is established as a legal entity

Wellcome Connecting Science grant - life sciences in schools

Open letter to support ELIXIR Membership by Romanian Scientists

2020
ELIXIR webinars for Romanian scientists

Beer & Science (satellite & self-standing meetings)

RSBI's first proposal to adopt a national strategy proposal for capacity building in bioinformatics (RoEduBioinfo)

2021
Joint UEFSCIDI- RSBI COVID 19 Information Portal

Covid19 Data Portal - data federation efforts

RSBI presents the case for ELIXIR Membership to the Romanian Government

RSBI proposal to include bioinformatics in the Research Roadmap

2022
Kickoff of the RSBI - ELIXIR Norway bilateral project

RSBI advocated for the genomic data federation, Romania's participation in European infrastructures, and political commitment to the 1 Million Genomes initiative

RSBI and SRGM led a consortium of 17 universities to create the Romanian Bioinformatics Cluster (CRB)

Exploratory work on federating human data within FEGA.

Exploratory work on a national Galaxy instance

2023
1st RoBioinfo Conference, Bucharest

Increased collaboration with ELIXIR Hub and the Ministry for Research, Innovation, and Digitalisation for Romania's ELIXIR accession

Webinar: ELIXIR Norway - structure and activity

2024
2nd RoBioinfo Conference, Cluj-Napoca

Webinar: Research communities in ELIXIR (ELIXIR Hub)

CRB joins ELIXIR Steer project

Romania signs the 1 Million Genome declaration 2023
Political commitment to join ELIXIR (Observer) 2023
Romania joins ELIXIR (Observer) 2023
Romania joins GDI (Onboarding) 2023
Romania submits the formal request to join ELIXIR as a Member 2024

Progress Towards National Federated Bioinformatics

2019-2020: Early attempts bypassing ELIXIR

2021-present: Raise awareness on ELIXIR

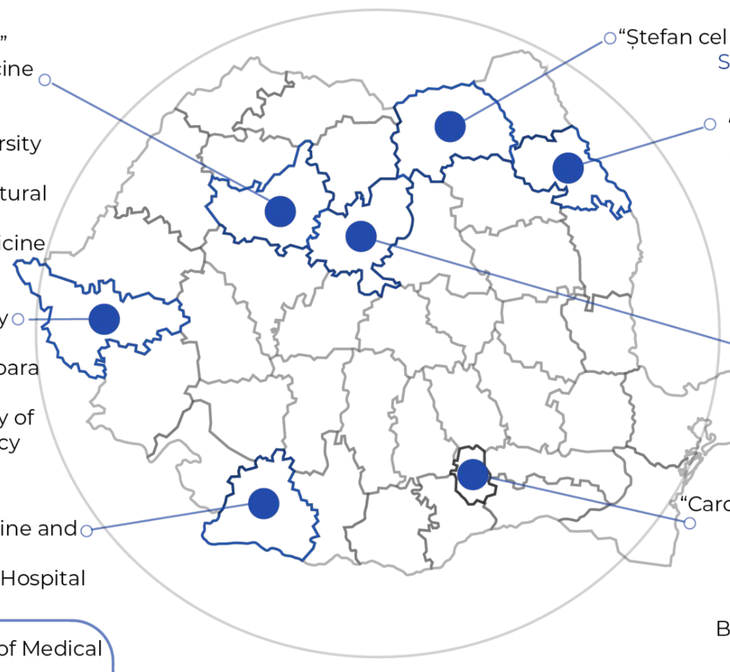
2022: CRB founded with commitment to ELIXIR interoperability

Spring 2023: Government considers joining ELIXIR

October 2023: ELIXIR Romania's Observership

April 2024: ELIXIR Membership request

Romanian Bioinformatics Cluster



- "Iuliu Hațieganu" University of Medicine and Pharmacy
- Babeș-Bolyai University
- University of Agricultural Sciences and Veterinary Medicine Cluj-Napoca
- Polytechnic University
- West University of Timisoara
- "Victor Babeș" University of Medicine and Pharmacy Timișoara
- University of Medicine and Pharmacy County Emergency Hospital Craiova
- Romanian Society of Medical Genetics
- Romanian Society of Bioinformatics
- "Ștefan cel Mare" University Suceava
- "Grigore T. Popa" University of Medicine and Pharmacy "Alexandru Ioan Cuza" University
- Regional Institute of Oncology Iași
- "George Emil Palade" University of Medicine, Pharmacy, Sciences, and Technology Târgu Mureș
- "Carol Davila" University of Medicine and Pharmacy
- University of Bucharest
- Biochemistry Institute of the Romanian Academy
- National Institute of Research and Development in Informatics Bucharest

Objectives

- Support national bioinformatics training
- Leverage existing training frameworks (e.g., Galaxy Training Network)
- Promote FAIR data principles
- Develop and maintain bioinformatics training resources
- Support data infrastructures for molecular data
- Implement novel data-sharing techniques
- Ensure uniform data access
- Facilitate easy access to bioinformatics tools with performance information

CRB Establisment

Founded by 17 universities and research centers

Partners: Romanian Society of Medical Genetics (SRGM) and RSBI

Future Directions

Data Initiatives
- Support national GDI Node
- Develop plant and pathogen data portals

Tools Development
- Advance European benchmarking framework
- Establish national tool repositories

Training Framework
- Decentralized contributions to European training resources
- Support national training programs

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